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Division: A2

ASSIGNMENT NO: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement: Implement a Java program to demonstrate inheritance and interfaces using suitable realworld examples.

Learning Objectives: To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory: Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Exercise:

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Program:

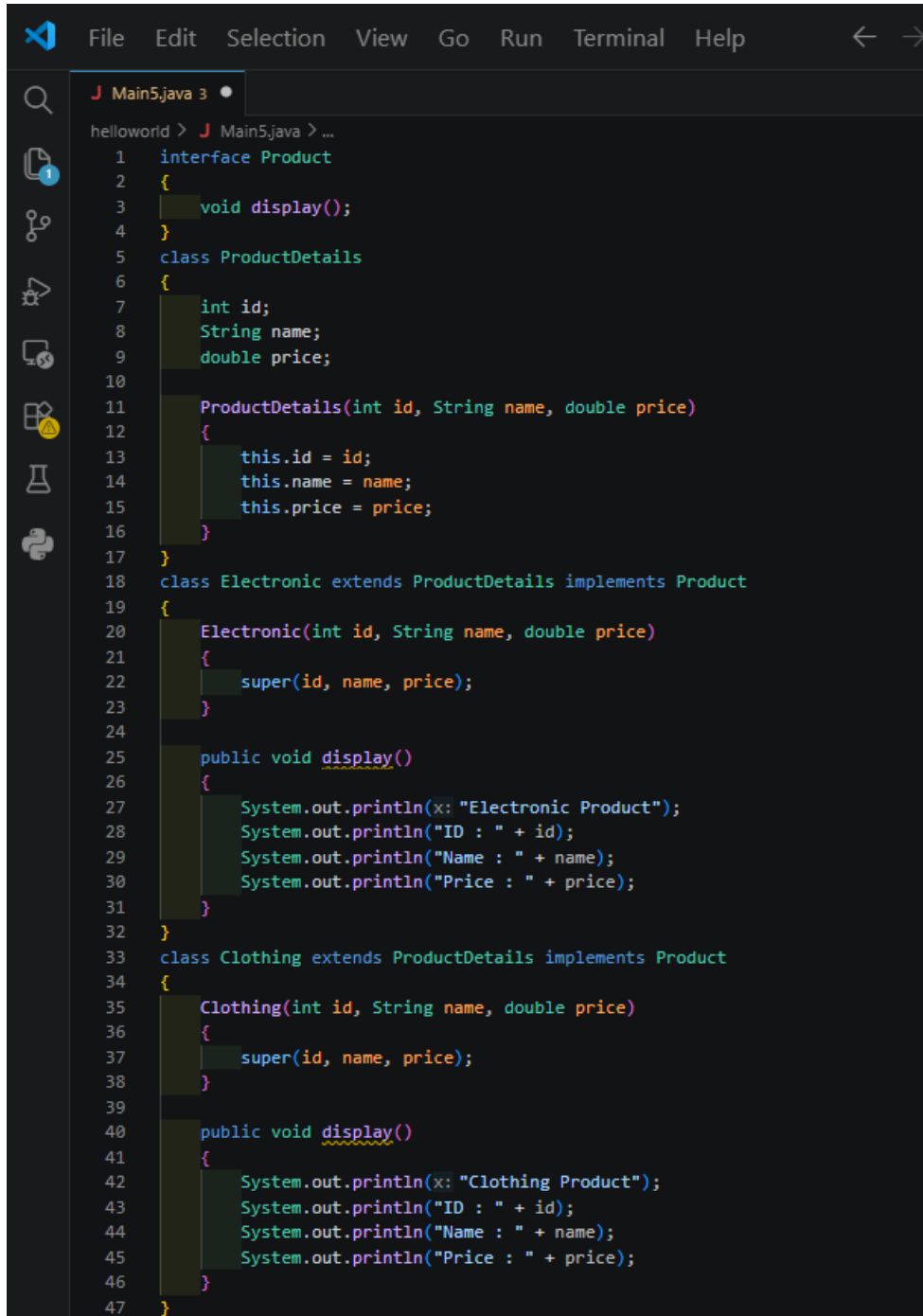
```
helloworld > J Main4.java > ...
1  import java.util.*;
2  class Shape
3  {
4      void area()
5      {
6          System.out.println("Area of Shape");
7      }
8  }
9  class Circle extends Shape
10 {
11     double radius;
12
13     Circle(double r)
14     {
15         radius = r;
16     }
17
18     void area()
19     {
20         System.out.println("Area of Circle = " + (3.14 * radius * radius));
21     }
22 }
23 class Rectangle extends Shape
24 {
25     int length, breadth;
26
27     Rectangle(int l, int b)
28     {
29         length = l;
30         breadth = b;
31     }
32
33     void area()
34     {
35         System.out.println("Area of Rectangle = " + (length * breadth));
36     }
37 }
38 public class Main4
39 {
40     Run main | Debug main | Run | Debug
41     public static void main(String args[])
42     {
43         Circle c = new Circle(5);
44         Rectangle r = new Rectangle(1, 4, 6);
45
46         c.area();
47         r.area();
48     }
49 }
```

Output:

```
File Edit Selection View Go R
PROBLEMS 14 OUTPUT DEBUG CONSOLE
● PS C:\Users\Ayush\projects> & 'C:\Program Files\Red Hat\Java\jdk-8.0.402\bin\java.exe' -cp 'C:\Users\Ayush\projects\Main4.jar' Main4
Area of Circle = 78.5
Area of Rectangle = 24
○ PS C:\Users\Ayush\projects>
```

2. Develop an E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Program:

A screenshot of a code editor window titled 'Main5.java 3'. The editor shows Java code for an E-commerce product system. It includes an interface 'Product' with a 'display()' method. Two classes, 'ProductDetails' and 'Electronic', inherit from 'ProductDetails' and implement the 'Product' interface. 'ProductDetails' has attributes 'id', 'name', and 'price'. 'Electronic' and 'Clothing' classes inherit from 'ProductDetails' and implement the 'display()' method. The code is as follows:

```
1 interface Product
2 {
3     void display();
4 }
5 class ProductDetails
6 {
7     int id;
8     String name;
9     double price;
10
11     ProductDetails(int id, String name, double price)
12     {
13         this.id = id;
14         this.name = name;
15         this.price = price;
16     }
17 }
18 class Electronic extends ProductDetails implements Product
19 {
20     Electronic(int id, String name, double price)
21     {
22         super(id, name, price);
23     }
24
25     public void display()
26     {
27         System.out.println("Electronic Product");
28         System.out.println("ID : " + id);
29         System.out.println("Name : " + name);
30         System.out.println("Price : " + price);
31     }
32 }
33 class Clothing extends ProductDetails implements Product
34 {
35     Clothing(int id, String name, double price)
36     {
37         super(id, name, price);
38     }
39
40     public void display()
41     {
42         System.out.println("Clothing Product");
43         System.out.println("ID : " + id);
44         System.out.println("Name : " + name);
45         System.out.println("Price : " + price);
46     }
47 }
```

```

48 class Grocery extends ProductDetails implements Product
49 {
50     Grocery(int id, String name, double price)
51     {
52         super(id, name, price);
53     }
54
55     public void display()
56     {
57         System.out.println("Grocery Product");
58         System.out.println("ID : " + id);
59         System.out.println("Name : " + name);
60         System.out.println("Price : " + price);
61     }
62 }
63 public class Main5
64 {
65     Run main | Debug main | Run | Debug
66     public static void main(String args[])
67     {
68         Electronic e = new Electronic(id: 101, name: "Laptop", price: 55000);
69         Clothing c = new Clothing(id: 102, name: "T-Shirt", price: 1200);
70         Grocery g = new Grocery(id: 103, name: "Rice", price: 900);
71
72         e.display();
73         System.out.println();
74
75         c.display();
76         System.out.println();
77
78         g.display();
79     }

```

Output:

```

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PROBLEMS 17 OUTPUT DEBUG CONSOLE
PS C:\Users\Ayush\projects> & 'C:\
Price : 55000.0

Clothing Product
ID : 102
Name : T-Shirt
Price : 1200.0

Grocery Product
ID : 103
Name : Rice
Price : 900.0
PS C:\Users\Ayush\projects>

```